

Unai Gurbindo Gonzalez

Data Scientist & AI Engineer | PhD Researcher in Trustworthy AI

Uncertainty Quantification | Reliable Machine Learning | OOD Detection | Responsible AI

Barcelona, Spain | unai.gurbindo@gmail.com | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#) | [ORCID](#)

Research Profile

Data Scientist and AI Engineer with a BSc in Data Science and research-engineering experience across reliable machine learning, computer vision and biomedical data analysis. Currently pursuing a PhD at the Barcelona Supercomputing Center (BSC-CNS) and Universitat Politècnica de Catalunya (UPC), focused on uncertainty quantification, out-of-distribution detection and machine-learning reliability under distribution shift. Experienced in building reproducible Python and PyTorch pipelines for model evaluation, uncertainty, calibration and robustness analysis. Co-first author of an ICML 2026 position paper and first author of an IJCNN 2025 paper.

Research Interests

Uncertainty quantification; out-of-distribution detection; distribution shift; calibration; domain uncertainty and deployment-domain validity; reliable machine learning; responsible AI.

Education

PhD in Artificial Intelligence, ongoing

Nov 2025 – Expected May 2029

Universitat Politècnica de Catalunya (UPC)

Barcelona, Spain

Doctoral research at BSC-CNS on uncertainty quantification, domain uncertainty and reliability under distribution shift. Supervisors: Axel Brando and Lluís Belanche. FI-STEP Predoctoral Fellow, funded by AGAUR.

Interuniversity MSc in Artificial Intelligence

Sep 2023 – Jan 2025

Universitat Politècnica de Catalunya / Universitat de Barcelona / Universitat Rovira i Virgili

Barcelona, Spain

GPA: 9.29/10. Thesis: *Image Generation for Improving the Robustness of Computer Vision Models in Autonomous Driving*. Resulted in a first-author publication at IJCNN 2025.

BSc in Data Science

Sep 2019 – Jun 2023

Public University of Navarre

Navarre, Spain

GPA: 8.95/10; ranked 1st in the graduating class. Thesis: *Characterisation of the Immune System in the Progression to ACLF*.

Publications

Peer-Reviewed Publications

- **Gurbindo Gonzalez, U.***, Aldoma, A.*, & Brando, A. *Position: Predictive Uncertainty Is Not Enough – Joint Distribution for Full Uncertainty Representation*. International Conference on Machine Learning (**ICML**), Position Paper Track, 2026. Presented at ICML 2026 in Seoul; proceedings forthcoming. *Equal contribution.
- **Gurbindo, U.**, Brando, A., Abella, J., & König, C. *Object Detection in Adverse Weather Conditions for Autonomous Vehicles Using Instruct Pix2Pix*. International Joint Conference on Neural Networks (**IJCNN**), 2025, pp. 1–8. [doi:10.1109/IJCNN64981.2025.11228623](https://doi.org/10.1109/IJCNN64981.2025.11228623).

Conference Poster & Published Abstract

- Palomino-Echeverria, S., Huergo, E., Uson, E. M., Sanchez, C., Lagani, V., Kiani, N., Planell, N., Trebicka, J., **Gurbindo, U.**, Farias, A. Q., Caraceni, P., Rautou, P.-E., & Gomez-Cabrero, D. *Robust Identification of Patient Subgroups in Acute Decompensated Cirrhosis*. Poster presentation at **EASL Congress 2023**, Vienna, Austria, SAT-337. Abstract published in *Journal of Hepatology*, 78(S1), S290, 2023. [doi:10.1016/S0168-8278\(23\)00877-2](https://doi.org/10.1016/S0168-8278(23)00877-2).

Submitted Manuscript

- **Gurbindo Gonzalez, U.**, Brando, A., & Valls, P. *Stop Footnoting Ethics in AI Documentation: Use Care Cards*. Submitted manuscript.

Research Experience

PhD Researcher in Trustworthy AI

Nov 2025 – Present

Barcelona Supercomputing Center & Universitat Politècnica de Catalunya

Barcelona, Spain

Supervisors: Axel Brando and Lluís Belanche.

- Formalise and empirically investigate *domain uncertainty* as a reliability problem distinct from predictive uncertainty, connecting uncertainty quantification with OOD detection and deployment-domain validity.
- Design experiments to study when predictive uncertainty becomes unreliable under distribution shift, including settings with ambiguous ID samples, corrupted inputs and OOD data.
- Develop reproducible Python and PyTorch workflows for comparing uncertainty estimation, calibration, OOD detection and robustness methods across models and evaluation settings.

Junior Research Engineer in Robust AI

Barcelona Supercomputing Center

Supervisor: Axel Brando.

Jan 2025 – Nov 2025

Barcelona, Spain

- Built reproducible Python and PyTorch pipelines to evaluate deep ensembles across ID, OOD, corrupted-data and mixed-distribution test settings.
- Compared uncertainty, calibration and robustness behaviour under distribution shift, analysing failure cases in which ambiguous ID samples and OOD inputs produced overlapping uncertainty signals.
- Designed empirical analyses and visualisations supporting a co-first-authored ICML position paper on joint uncertainty representation.

MSc Thesis Researcher in Autonomous Driving

Barcelona Supercomputing Center

Supervisors: Axel Brando, Caroline König and Jaume Abella.

Jul 2024 – Jan 2025

Barcelona, Spain

- Developed a generative data-augmentation pipeline using InstructPix2Pix to improve object-detection robustness under adverse-weather distribution shifts.
- Evaluated YOLOv10 and Faster R-CNN on CARLA, BDD100K and ACDC, comparing performance across clean, synthetic and real adverse-weather conditions.
- Produced the experimental work leading to a first-author publication at IJCNN 2025.

Bioinformatics Research Intern and Bachelor Thesis Researcher

Navarrabiomed, Translational Bioinformatics Unit

Biomedical research centre.

Sep 2022 – Jun 2023

Navarre, Spain

- Analysed clinical and single-cell biomedical data to identify immune-system patterns associated with acute-on-chronic liver failure.
- Applied statistical analysis, patient stratification and data visualisation in a translational bioinformatics project, contributing as a co-author to a poster presented at EASL Congress 2023, with its abstract published in *Journal of Hepatology*.

Technical Skills

Data Science & Programming: Python (primary), SQL, R, Bash, MATLAB; statistical analysis, data processing and visualisation

Machine Learning & Deep Learning: PyTorch, TensorFlow/Keras, scikit-learn, NumPy, pandas, SciPy, Hugging Face

Reliable ML & Computer Vision: uncertainty quantification, calibration, OOD detection, distribution-shift evaluation, corruption robustness, deep ensembles, object detection

Research Engineering: Git, Linux, Docker, CUDA/GPU, pytest, GitHub Actions, Weights & Biases, Jupyter, reproducible experimentation, LaTeX

Responsible AI: AI documentation, risk assessment, uncertainty communication, responsible deployment

Selected Presentations and Teaching

Poster Presenter, ICML 2026

International Conference on Machine Learning

Presented the co-first-authored position paper *Predictive Uncertainty Is Not Enough*.

Jul 2026

Seoul, South Korea

Paper Presenter, IJCNN 2025

International Joint Conference on Neural Networks

Presented the first-author work on generative augmentation for adverse-weather object detection.

2025

Rome, Italy

Instructor, Trustworthy AI

Chamber of Commerce of Terrassa

Delivered applied sessions for SMEs on uncertainty, risk assessment, AI documentation and responsible adoption.

Jan 2026 – Feb 2026

Terrassa, Spain

Project Presenter, European Researchers' Night

Public University of Navarre

Presented the UPNAvarra_Spain iGEM project to a public audience.

Sep 2022

Navarre, Spain

Awards, Funding and Languages

Awards & Funding: FI-STEP Predoctoral Fellowship, AGAUR – Government of Catalonia, awarded Sep 2025; highest course distinction (*Matrícula de Honor*) in 10 BSc courses and 2 MSc courses.

Languages: Basque – Native; Spanish – Native; English – B2 (certified), upper-intermediate working proficiency.